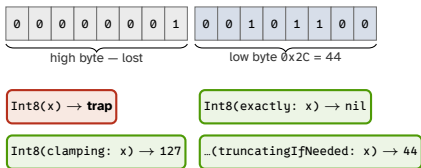


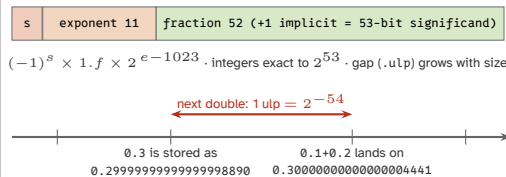
**In one line:** Swift integer arithmetic **traps** on overflow (in Debug *and* Release) — wrapping is opt-in with `&+ &- &*`; every lossy conversion has a named init that says *how* it loses. Double is binary IEEE 754, so decimal fractions are approximations: compare with a tolerance, and keep money in `Decimal` (built from strings) or integer minor units.

Let `x = 300 (0x012C)` into an `Int8`

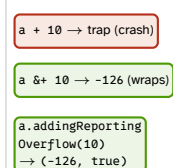


literal `Int8(300)` = compile error; a runtime value traps

Double = IEEE 754 binary64



`Int8(120) plus 10`



## Integers

- `Int/UInt` = word size: 64-bit on iPhone/Mac, but **32-bit on arm64\_32** (Apple Watch Series 4+). Use `Int` everywhere (count, indices); sized types for file/wire formats and C. `Int128/UInt128`: Swift 6 (SE-0425).
- `+ - *` **trap** on overflow in `-0none` and `-0` (only `-0unchecked` removes checks). Also trap: `/ 0`, `Int.min / -1`, `abs(Int.min)`, `UInt 0 - 1`. A trap is a crash, not an exception — no catch.
- `&+ &- &*`: wrap modulo  $2^n$  (hashes, checksums, ring counters). `&<&>`: shift count masked mod `bitWidth`; plain `<<` with a big count gives `0`, never traps.
- `addingReportingOverflow` (also subtracting, multiplied, divided)  $\rightarrow$  (partialValue, overflow); `multipliedFullWidth(by:)`  $\rightarrow$  (high, low).
- Conversions: `init(_:)` traps · `exactly`: optional · `clamping`: saturates · `truncatingIfNeeded`: low bits · `bitPattern`: reinterprets (`UInt8(bitPattern: -1) = 255`). Generic code: `FixedWidthInteger` (max, bitWidth, `&+`) refines `BinaryInteger`.

## Floating point

- `Float16` (11-bit significand,  $\approx 3$  digits; iOS 14+, not on Intel Macs) · `Float` (24-bit,  $\approx 7$  digits, integers exact only to 16 777 216) · `Double` (53-bit). Default to `Double`; `CGFloat` is `Double` on 64-bit and converts implicitly since Swift 5.5 (SE-0307).
- Float ops **never trap**: `1/0.0 = +inf`, `0/0.0 = NaN`. NaN is unequal to everything, itself included — test `x.isNaN`; NaNs poison sort, min, max. `-0.0` equals `0.0`.
- `Double.ulpOfOne = 2-52  $\approx$  2.2 × 10-16` (machine epsilon); `x.ulp` = gap at `x`; `nextUp` steps. Compare:  $|a - b| \leq \max(\text{abs}, \text{rel} \cdot \max(|a|, |b|))$ , or `swift-numeric` `isApproximatelyEqual(to:)`.
- `Int(2.99) = 2`, `Int(-2.99) = -2` (toward zero; traps on NaN/inf/out of range) — round first on purpose.

## Example

```
let big = 300
Int8(clamping: big)           // 127
Int8(truncatingIfNeeded: big) // 44
Int8(exactly: big)           // nil
(2.5).rounded(.toNearestOrEven) // 2.0 banker's
let bad = Decimal(0.1)       // literal goes through Double
let good = Decimal(string: "0.10")! // exact
(good * 3).formatted(.currency(code: "EUR")) // 0.30, localised
```

## Money: Decimal and its own caveats

- `Decimal` (Foundation, = `NSDecimal`): base-10, 128-bit mantissa ( $\approx 38$  digits), exponent  $-128 \dots 127$ . `0.1` is exact — *if* built from a string or integers.
- Float literal trap**: let `d: Decimal = 0.1` and `Decimal(0.1)` go through `Double` and can carry binary noise. Use `Decimal(string:)` or integer maths (`Decimal(10) / 100`).
- Still rounds (1/3); not `FloatingPoint` (no `.rounded()`) — use `NSDecimalRound(&out, &in, 2, .bankers)`. Software maths: slower than `Double`.
- `NSDecimalNumber` (class; rounding via a behaviour handler) by default **raises an ObjC exception** on divide-by-zero or overflow — a crash, not a throw.
- Or **integer minor units** (`Int64` cents; JPY has 0, KWD 3). On the wire: string or integer, never a JSON float.
- Display: `.currency(code:)`, `.precision(.fractionLength(2))` — not `String(format: "%.2f")` (`Double`, no locale).

## Rounding rules

| FloatingPointRoundingRule             | 2.5   | -2.5    | use                    |
|---------------------------------------|-------|---------|------------------------|
| <code>.toNearestOrAwayFromZero</code> | 3     | -3      | <code>rounded()</code> |
| <code>.toNearestOrEven</code>         | 2     | -2      | banker's, no bias      |
| <code>.up / .down</code>              | 3 / 2 | -2 / -3 | ceil / floor           |
| <code>.towardZero</code>              | 2     | -2      | <code>= Int(x)</code>  |

## Traps

- ! “Swift wraps like C” — no: it **crashes**. `reduce(0, +)` on big counts traps; use a wider type or reporting ops.
- ! `UInt` for “can’t be negative” counts: `a - b` traps when `b > a`. Prefer `Int`.
- ! `0.1 + 0.2` vs `0.3`: unequal (one ulp apart). No equality tests on computed floats; no `Double` for money; IDs in `Float` lose digits past  $2^{24}$ .

## Remember

**“Plus traps, ampersand wraps, reporting tells.”** Conversions: *trap* · *exactly* · *clamping* · *truncating* · *bitPattern*. Money: strings in, `Decimal`/cents inside, `FormatStyle` out.